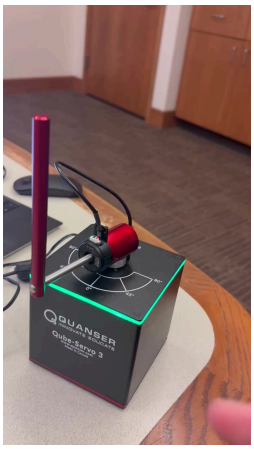
Project Portfolio

Work in progress

Capstone

For Joel's senior capstone class he is furthering Dr. Anchal Sareen's research on Unmanned Underwater Vehicle steering. Working on fluid dynamics, control, design, laboratory testing, and data analysis.

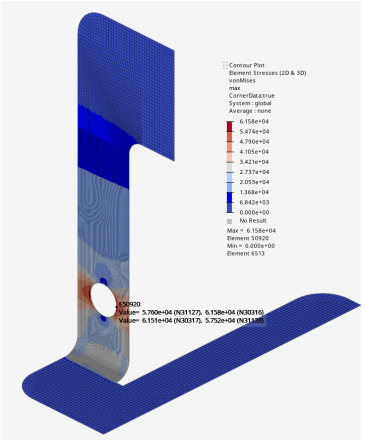


Testing controller on Quanser Qube-Servo 3

Controls

Performed design of PID controllers in time and frequency domain. Designed full-state feedback and compensated full-state feedback controllers. Pictured full-state controller was designed to stabilize an inverted pendulum system with a full-state feedback controller built on Simulink.

Details/Documentation

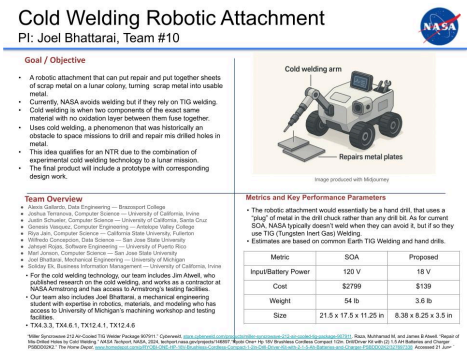


Von Mises stress contour on a bookstop

Finite Element Analysis

Analyzed a variety of shapes in HyperMesh and OptiStruct. Leveraging 2D and 3D elements, convergence studies, and buckling analysis, to make design recommendations on a variety of systems.

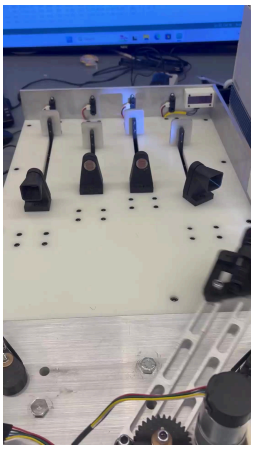
Details/Documentation



Quad chart for cold welding attachment concept

Principal Investigator

Proposed an new technology concept that was selected by a team of NASA engineers and stakeholders. Led a team of interdisciplinary students to a full technical proposal including a NASA New Technology Report. Also served on a NASA review board for three other proposals as the primary reviewer on the panel.

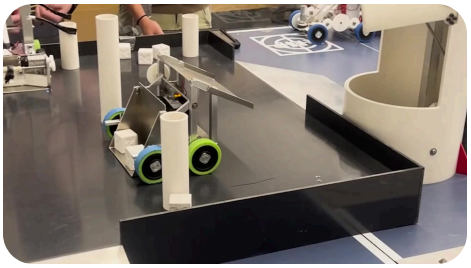


Linkage playing "game" to shine light on four different targets based on block positions.

Linkage Development

Leveraged MSC Adams for multibody dynamic simulation to model the linkage system to hit all four targets. Designed transmission, links, and hardstops for a robust solution. Performed PID tuning to ensure the speed of performance and fast settling.

Details/Documentation

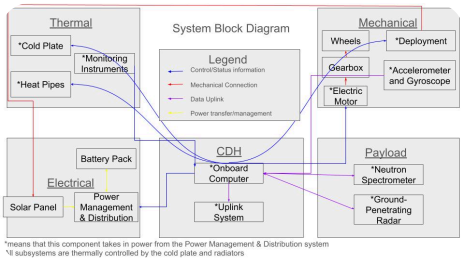


Robotic Machine Player confirming points for our team by delivering cubes into the delivery location

RMP Development

Developed CAD and design skills to complete multiple different tasks in a high-pressure competition. Designed robust transmission and powertrain systems. Executed analytical statics analysis to ensure the functionality of multiple functions simultaneously.

Details/Documentation



Full rover system block diagram

Lead Systems Engineer

Led a group of engineering students to a PDR for a lunar rover mission. Coordinated with the Science and Programmatics team on developing system-level requirements, and flowed these requirements down to Mechanical, Electrical, and Thermal requirements. Managed all interfacing between systems as well as protocol development.